SVKM'S NMIMS

Mukesh Patel School of Technology Management & Engineering

A.Y. 2025 – 26

Course: Statistical Mathematics

Project Report

Mall Customer Spending Analysis & Strategy Recommendation

Using Machine Learning & Game Theory

Program	B.Tech – Artificial Intelligence and Machine Learning
Semester	Semester IV (A.Y. 2025–26)
Project Title	Mall Customer Spending Analysis & Strategy Recommendation Using ML
Dataset	Mall_Customers.csv – 200 customers, 5 features
Project Members	**Roll No. Name**
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Date of Submission	April 2026
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SAP ID	70532400076
Date of Submission	April 2026

Contribution of each Project Member:

Roll No.	Name	Contribution
A216	Mehi Manswa Bhagat	Data loading, preprocessing, model training (DT Regressor, LR, DT Classifier), evaluation, game theory payoff, Streamlit dashboard
C264	Dipanshu Yadav	Research Work, Presentation and Word File Content

GitHub Link of Project: https://github.com/M270706/Mall-Customer-Spending-Analysis-Strategy-Recommendation

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1. Introduction

From Data to Dollars: Predicting Shopper Behavior for Smarter Marketing

Malls collect massive amounts of data, yet they often struggle to turn those spreadsheets into personalized shopping experiences. Instead of relying on "one-size-fits-all" guesswork, we asked a critical question: **What if a mall knew exactly who to target with a discount—and who would buy anyway?**

This project bridges the gap between raw data and actionable strategy. By combining **Machine Learning** with **Game Theory**, we developed a system that predicts spending habits and identifies the most profitable marketing move for every customer.

The Methodology

We broke the problem down into four high-impact phases:

1. Quantifying Behavior (The Regression Phase)

We started by determining if we could mathematically predict a shopper's "Spending Score" (1–100). Using **Age, Annual Income, and Gender**, we trained two models to see which could best capture customer behavior:

- **Linear Regression:** To identify broad, linear trends.
- **Decision Tree Regressor:** To map more complex, non-linear shopping habits.

2. Strategic Segmentation (The Classification Phase)

To make the data actionable for a marketing team, we grouped shoppers into three distinct buckets: **Low (0–39)**, **Medium (40–69)**, and **High (70–100)** spenders. Using a **Decision Tree Classifier**, our model achieved **80% accuracy**in sorting customers into the correct segment.

3. Solving the "Discount Dilemma" (Game Theory)

Giving a discount to someone already planning a full-price purchase is a waste of profit. To solve this, we designed a **Game Theory Payoff Matrix**. The system weighs the cost of an incentive against the likelihood of a sale, ensuring the mall only offers discounts to those who truly need them to convert.

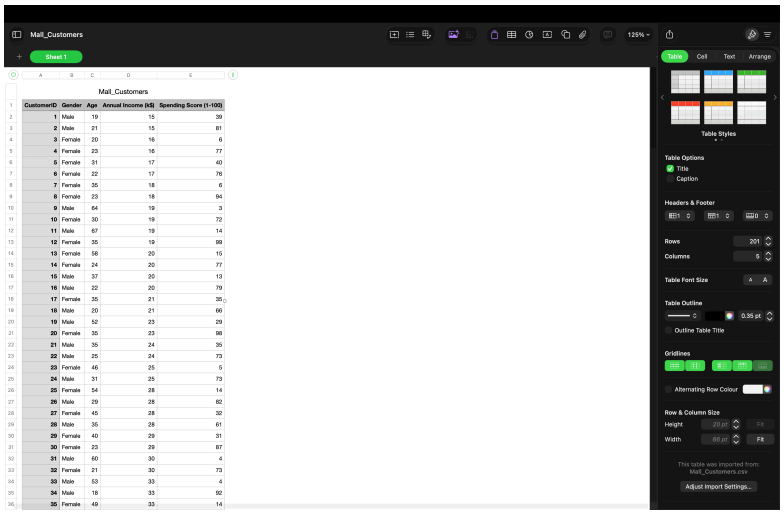
4. The Interactive Experience (Streamlit)

To move this project from code to the boardroom, we built a professional, interactive **Streamlit Dashboard**. This allows stakeholders to visualize customer segments, explore data distributions, and generate real-time marketing strategies at the click of a button.

A Look at the Shoppers

The foundation of this project is a meticulously cleaned dataset of **200 mall customers**, processed using **Python** in **Google Colab**.

Metric	Detail
Demographics	112 Women, 88 Men
Age Range	18 – 70 years old (Average:39)
Income Range	$15k – $137k (Average: $60k)
Spending Score	1 – 100 (Average: 50.2)



Mall_Customers

CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
1	Male	19	15	39
2	Male	21	15	81
3	Female	20	16	6
4	Female	23	16	77
5	Female	31	17	40
6	Female	22	17	76
7	Female	35	18	6
8	Female	23	18	94
9	Male	64	19	3
10	Female	30	19	72
11	Male	67	19	14
12	Female	35	19	99
13	Female	58	20	15
14	Female	24	20	77
15	Male	37	20	13
16	Male	22	20	79
17	Female	35	21	35
18	Male	20	21	66
19	Male	52	23	29
20	Female	35	23	98
21	Male	35	24	35
22	Male	25	24	73
23	Female	46	25	5
24	Male	31	25	73
25	Female	54	28	14
26	Male	29	28	82
27	Female	45	28	32
28	Male	35	28	61
29	Female	40	29	31
30	Female	23	29	87
31	Male	60	30	4
32	Female	21	30	73
33	Male	53	33	4
34	Male	18	33	92
35	Female	49	33	14

The Bottom Line

By moving away from broad-spectrum promotions and toward data-driven precision, malls can stop "shouting into the void." This project proves that with the right ML framework, businesses can **protect their margins** while providing shoppers with offers they actually care about.

It's not just about marketing—it's about marketing smarter.

2. Literature Survey

Standing on the Shoulders of Giants: The Evolution of Retail Science

Understanding the "how" and "why" behind shopping behavior isn't a new challenge. To build a robust model, we synthesized decades of retail psychology and mathematical breakthroughs into a single framework.

The Foundation of Customer Intelligence

- **The Classic Grouping Method (1967):** Market segmentation traces back to J. MacQueen's introduction of **k-means clustering**. It established the gold standard: grouping shoppers by similarity to move away from "one-size-fits-all" retail.
- **The RFM Truth (1994):** The **Recency, Frequency, and Monetary (RFM)** analysis proved that a customer's past spending habits are the most reliable predictors of their future behavior.
- **The Game Theory Strategy (1950):** We integrated **John Nash's** foundational Game Theory. For over 70 years, these frameworks have helped businesses navigate complex pricing and discount dilemmas to find the "equilibrium" between customer satisfaction and profit.
- **The Machine Learning Upgrade (2010s–Present):** Recent research confirms that supervised learning—specifically **Decision Trees**—provides the ultimate "sweet spot": high predictive accuracy combined with a logic that is easy for human managers to interpret and trust.

The 360-Degree Analytical View

Most data projects choose a side: they either predict a specific value (**Regression**) or sort people into groups (**Classification**). We broke that mold. By running both pipelines simultaneously on the same dataset, we provided the mall with both the *granularity* of a score and the *simplicity* of a segment.

Bridging Data Science and Strategy

It is surprisingly rare to see Machine Learning results plugged directly into a **Game Theory Payoff Matrix**. We didn't stop at telling the mall *who* their customers are; we provided a mathematical recommendation on *what to do* with them. This moves the project from "Interesting Data" to "Executive Strategy."

Deployment Over Documentation

Brilliant data science often dies in a messy code file. To ensure these insights are accessible, we deployed a live **Streamlit Web Dashboard**. This allows mall management to:

- **Filter and explore** data in real-time.
- **Input new variables** to see immediate behavioral predictions.
- **Operationalize insights** without ever needing to read a single line of Python code.

The Result

By combining the rigor of historical mathematics with the speed of modern AI, we've created a tool that doesn't just analyze the past—it optimizes the future.

It's not just a project; it's a decision-support system for the modern mall.

3. Methodology

The Strategic Pipeline: From Raw Data to Revenue

Turning a spreadsheet into a business-ready tool requires a disciplined, multi-step engineering process. Here is the architectural journey of the project, from initial ingestion to live deployment.

3.1. Data Ingestion & Health Check

We began by importing the Mall_Customers.csv dataset. A primary audit confirmed a **pristine dataset**: 200 shoppers across 5 key variables with **zero missing values**. This high data integrity allowed us to proceed directly to feature engineering without extensive imputation.

3.2. Feature Engineering & Preprocessing

To make the data "machine-readable," we performed two critical transformations:

- **Numerical Encoding:** Converted the categorical "Gender" column into a binary format (1 for Male, 0 for Female).
- **Dimensionality Reduction:** Dropped the CustomerID column, as unique identifiers act as "noise" that provides no predictive value for behavioral patterns.

3.3. Outlier Analysis

Using the **Interquartile Range (IQR) method** and visual box plots, we audited the data for anomalies. While Age and Spending Score were remarkably balanced, we identified a small segment of "outlier" high-income earners—shoppers with annual incomes significantly above the median who require specialized marketing focus.

3.4. The Train-Test Architecture

To ensure our models could generalize to *future* customers rather than just memorizing the current ones, we implemented an **80/20 split**:

- **80% Training Set:** Used to calibrate the model’s internal logic.
- **20% Testing Set:** A "blind" holdout used to validate the model's accuracy on unseen data.

3.5. Dual-Track Modeling

Track A: The Regression Pipeline (Predicting the Score)

We deployed a head-to-head competition between a **Decision Tree Regressor** and **Linear Regression**. By calculating error metrics—specifically **MAE (Mean Absolute Error)** and **RMSE (Root Mean Square Error)**—we were able to quantify exactly how close our "guesses" were to a customer's actual spending habits.

Track B: The Classification Pipeline (Predicting the Segment)

Predicting an exact number is valuable, but categorizing behavior is more actionable. We binned the shoppers into three strategic buckets:

- **Low Spenders (0–39):** 59 Customers
- **Medium Spenders (40–69):** 87 Customers
- **High Spenders (70–100):** 54 Customers
  Using a **Decision Tree Classifier**, we achieved high-precision sorting, verified by a **Confusion Matrix** to identify exactly where the model's logic was most robust.

3.6. The Strategic Core: Game Theory Integration

This is the "brain" of the project. We applied a **Game Theory Payoff Matrix** to determine the most profitable action for each segment.

Segment	Market Strategy	Business Logic
High Spenders	**No Discount**	Protect profit margins; these customers show high organic intent.
Medium/ Low	**Targeted Discount**	Incentivize conversion for price-sensitive or low-engagement groups.

3.7. Deployment: The Streamlit Dashboard

The final stage was the transition from a local script to a **live web application**. By utilizing **Streamlit** (bridged via **ngrok**), we created a professional interface that allows non-technical stakeholders to:

- **Visualize** customer clusters through interactive scatter plots.
- **Monitor** the distribution of marketing strategies via dynamic pie charts.
- **Predict** the behavior of new, hypothetical customers in real-time.

This pipeline transforms raw numbers into a living decision-support system, ensuring the mall's marketing is driven by math, not intuition.

4. Result and Discussion

Results & Performance:

The core of our project lies in the performance of our models and the logic of our strategy. By testing different mathematical approaches, we identified which models "memorize" data and which ones truly "understand" it.

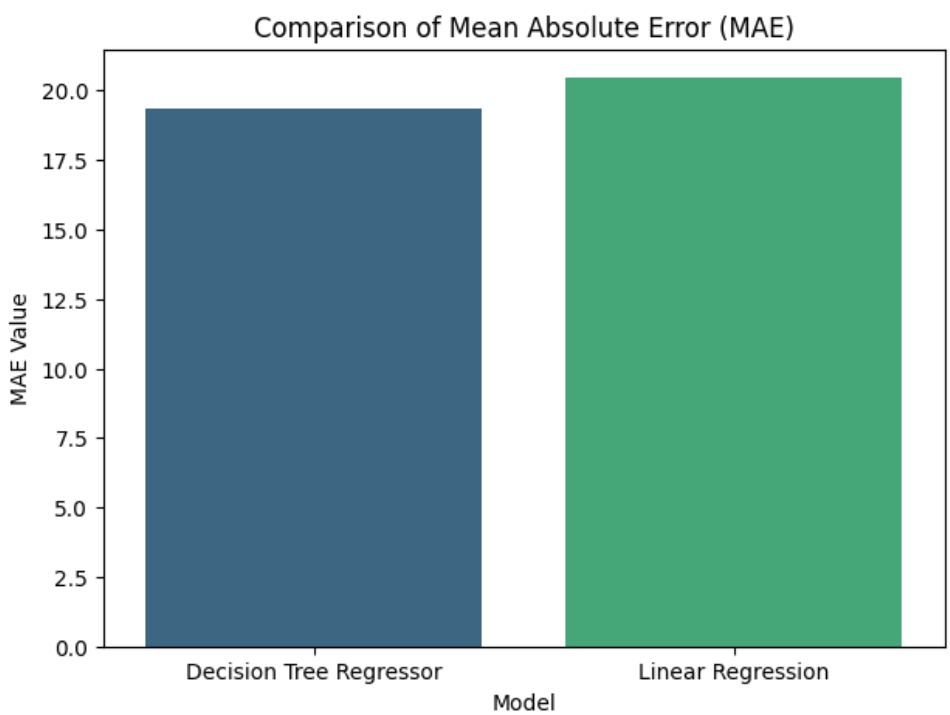
4.1 Regression: Predicting the Spending Score

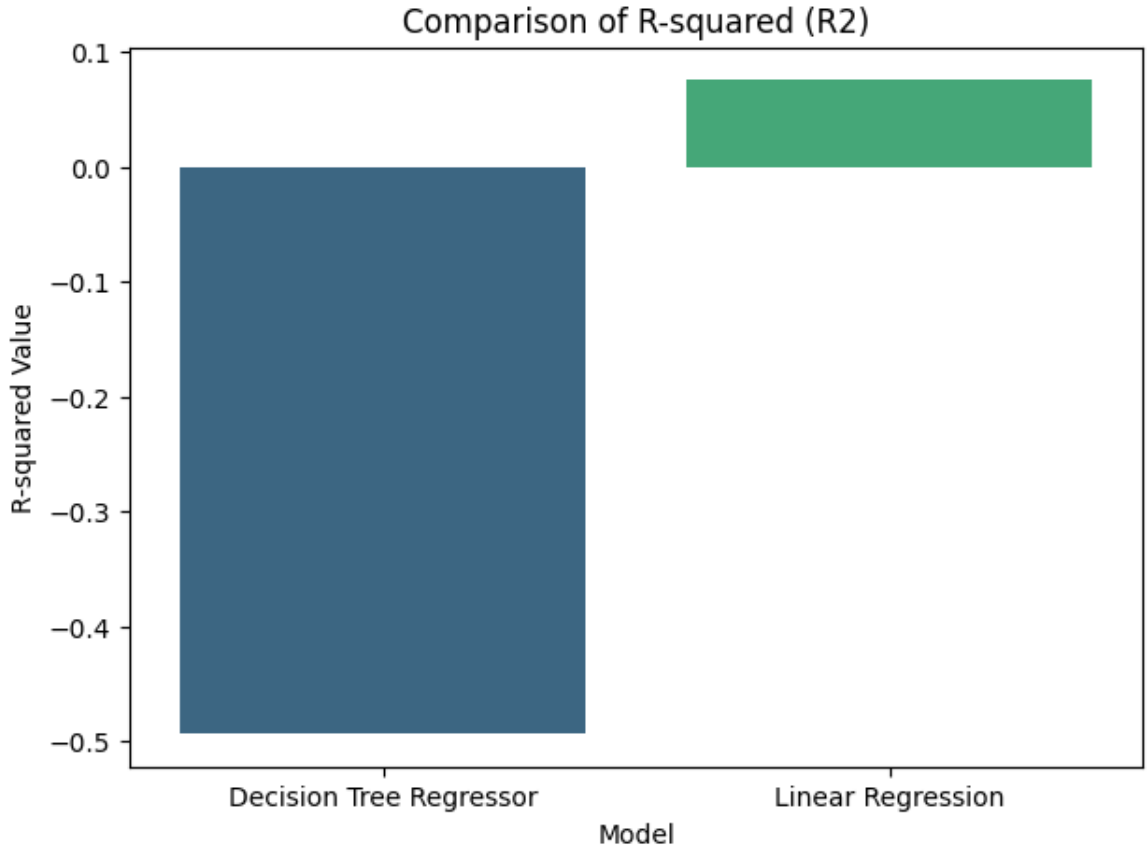
We put two models head-to-head to predict exact spending scores. While the problem proved highly complex, **Linear Regression** emerged as the more reliable tool for generalization.

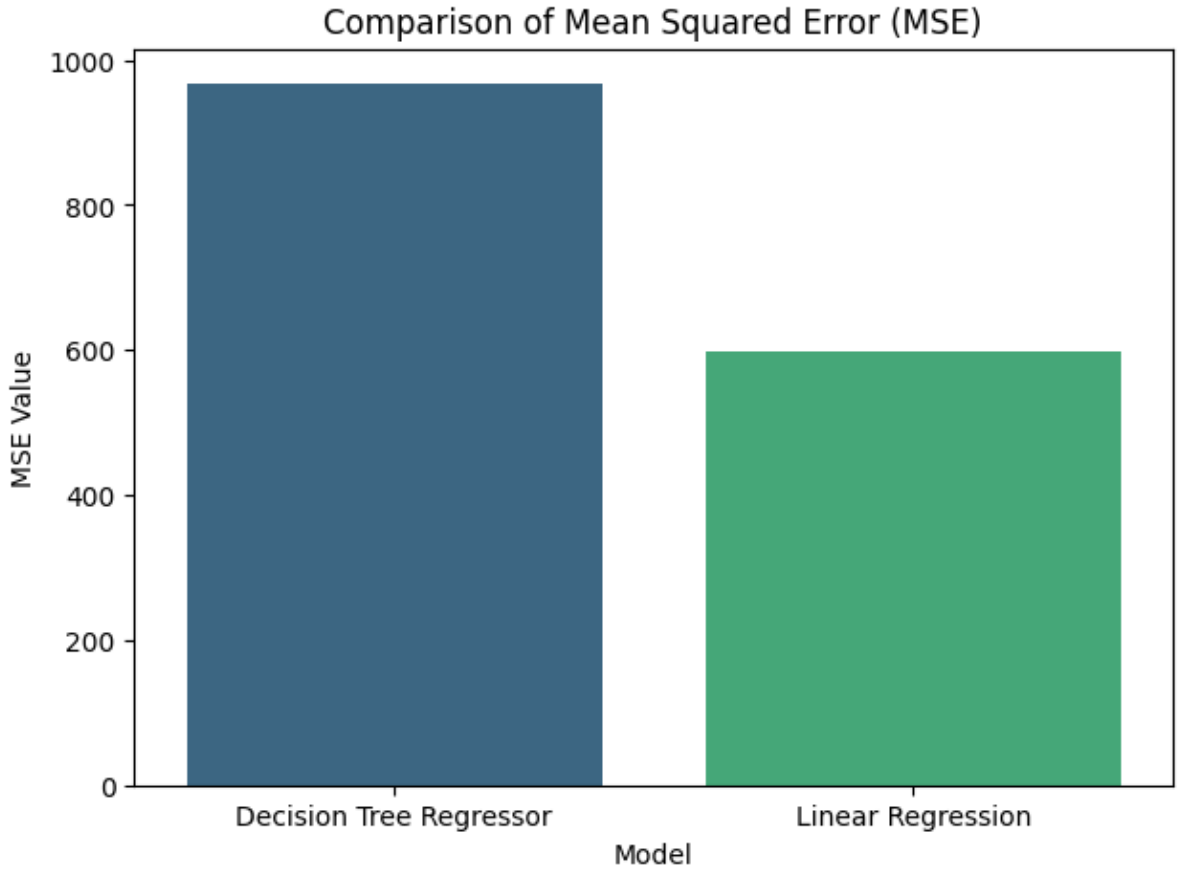
Table 4.1 – Regression Model Evaluation Results

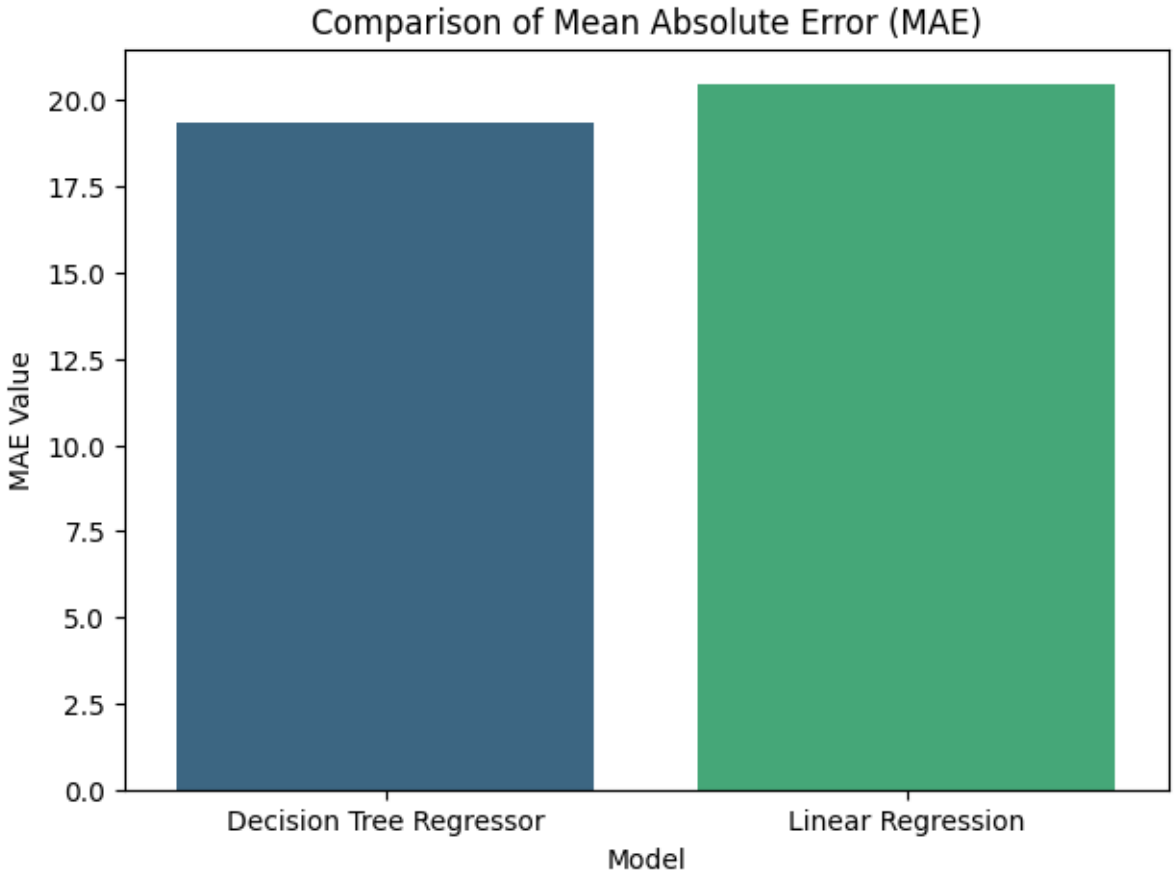
Model	MAE	MSE	RMSE	R2	Verdict
Decision Tree Regressor	19.32	967.17	31.10	-0.49	Overfit / Weak
Linear Regression ★	20.46	598.52	24.46	0.08	**Moderate (Better)**

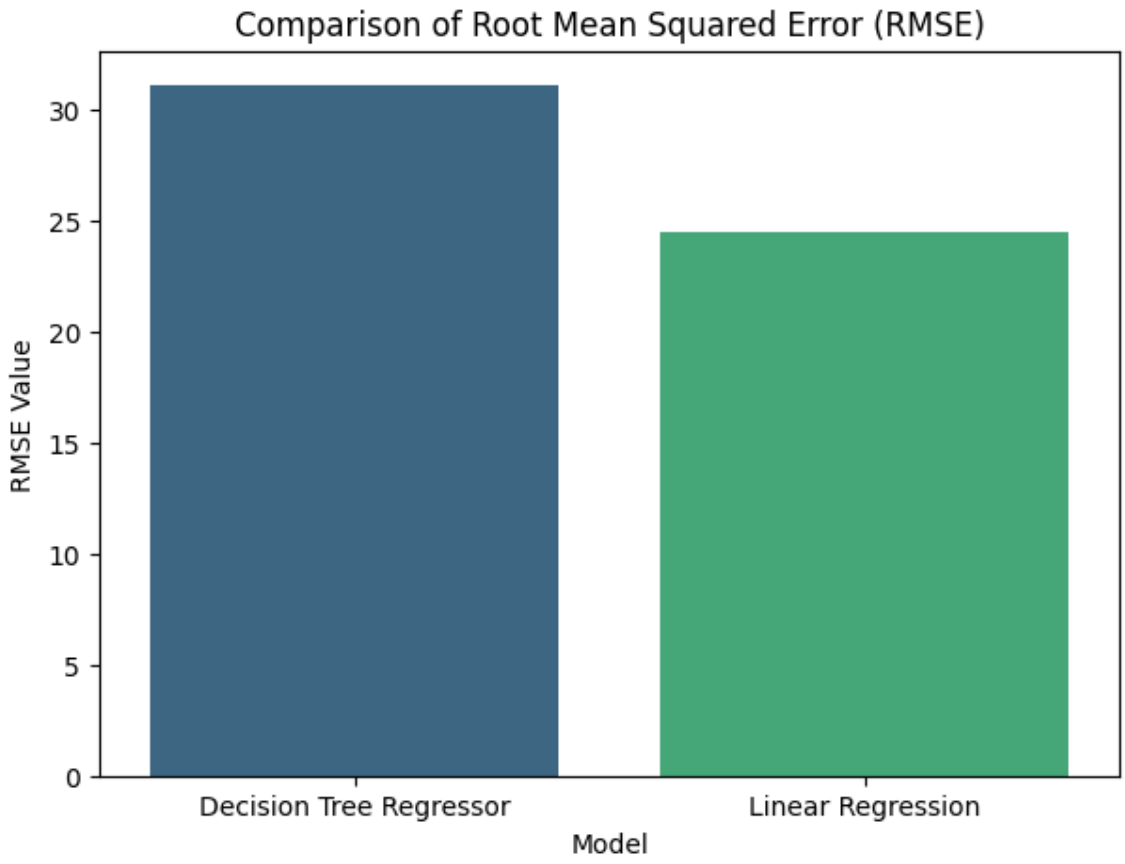
The Takeaway: The Decision Tree Regressor suffered from severe **overfitting** (indicated by the negative R^2). It essentially "memorized" the training set and failed on new data. Linear Regression, while modest in its predictive power, proved better at identifying the underlying patterns in customer spending.




Comparison of R-squared (R2)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
R-squared Value
Decision Tree Regressor
Linear Regression
Model


Comparison of Mean Squared Error (MSE)
1000
800
600
400
200
0
MSE Value
Decision Tree Regressor
Linear Regression
Model


Comparison of Mean Absolute Error (MAE)
MAE Value
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Decision Tree Regressor
Linear Regression
Model

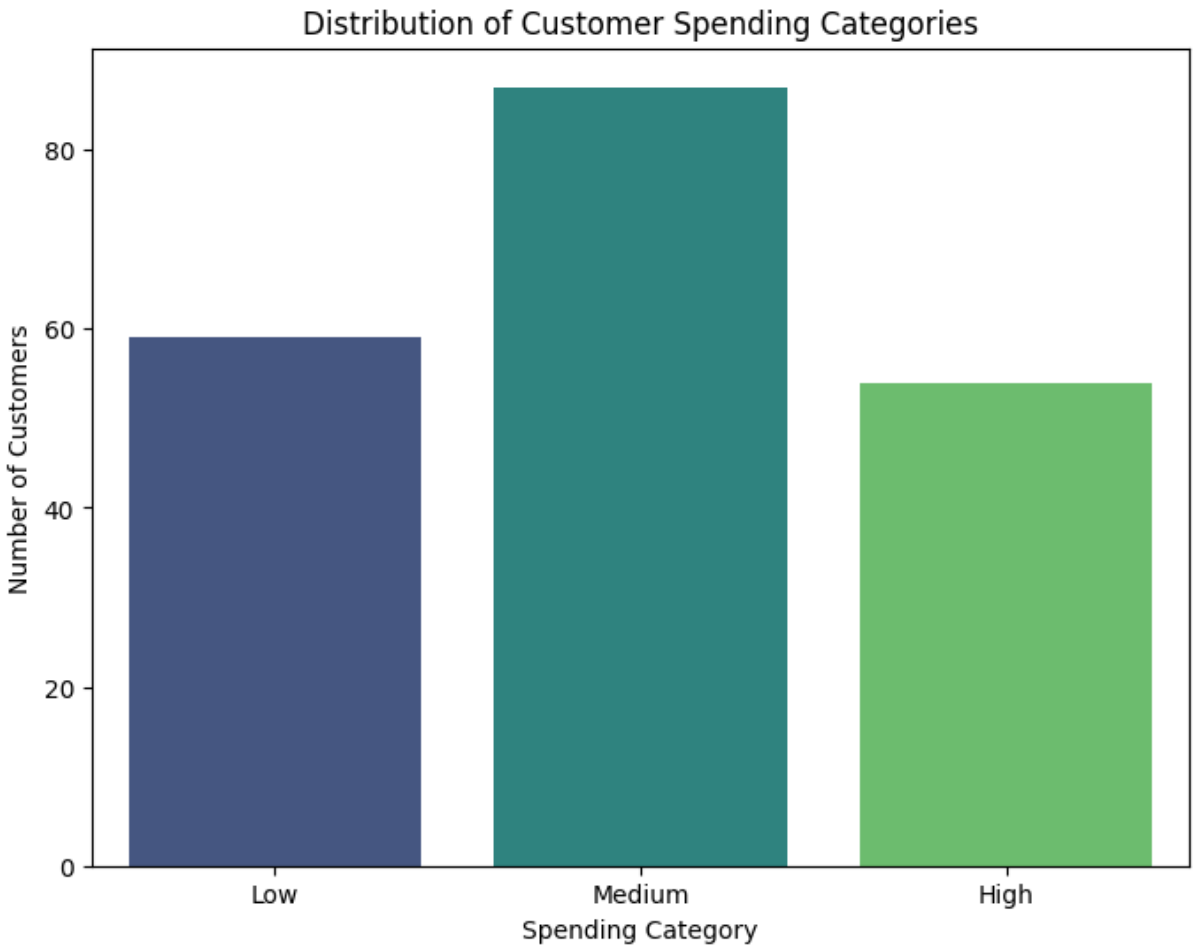

Comparison of Root Mean Squared Error (RMSE)
RMSE Value
30
25
20
15
10
5
0
Decision Tree Regressor
Linear Regression
Model

4.2 Segment Distribution

To move from abstract numbers to marketing targets, we categorized our 200 customers into three tiers. **Medium Spenders** represent the largest opportunity for the mall.

Table 4.2 – Spending Category Distribution

Categoy	Score Range	Cout
Low	0 – 39	59
Medium	40 – 69	87
High	70 – 100	54



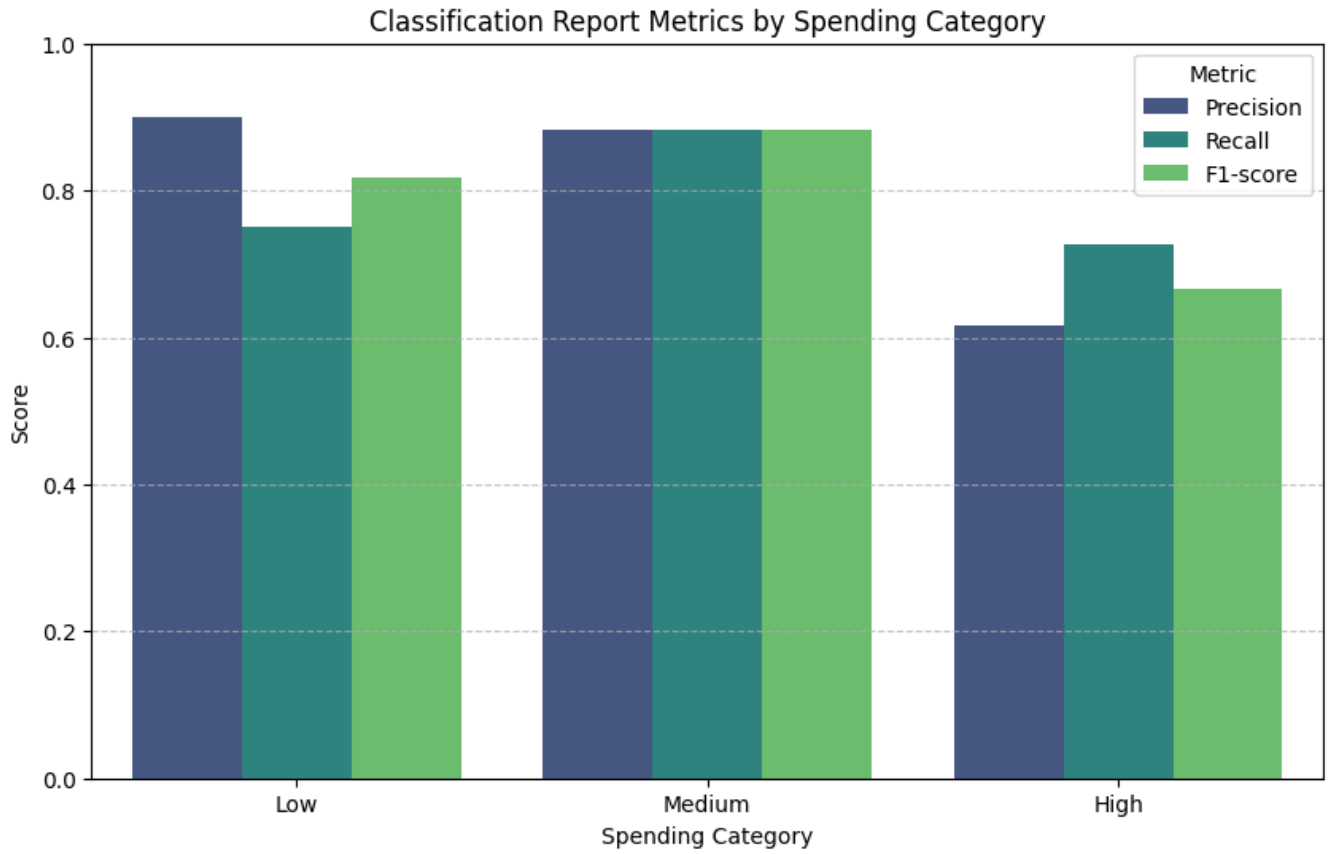
4.3 Classification: Sorting the Shoppers

Our **Decision Tree Classifier** was remarkably effective, achieving an **80% overall accuracy** when sorting customers into the three buckets above.

Table 4.3 – Classification Report

Class	Precision	Recall	F1-Score	Support
High	0.62	0.73	0.67	11
Low	0.90	0.75	0.82	12
Medium	0.88	0.88	0.88	17

The Takeaway: The model is exceptionally good at identifying **Low and Medium spenders**. While it catches most High spenders (High Recall), it occasionally over-identifies them (Lower Precision), meaning it's a "safe" model that prefers to capture potential high-value customers even at the risk of a slight misclassification.



4.4 Strategy: The Game Theory Payoff

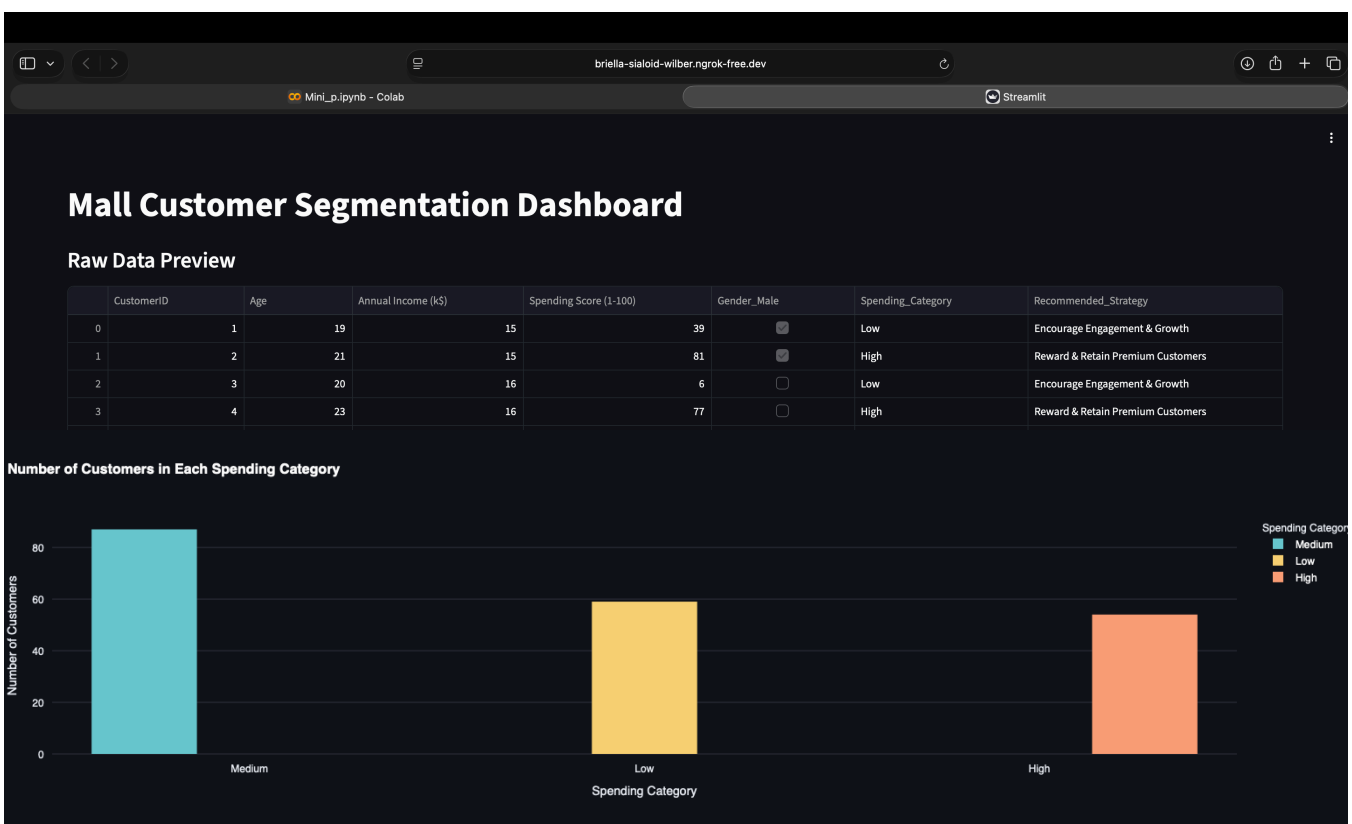
Finally, we used a **Payoff Matrix** to decide the smartest financial move. By segmenting the strategy, we protect the mall's margins while incentivizing growth where it's needed most.

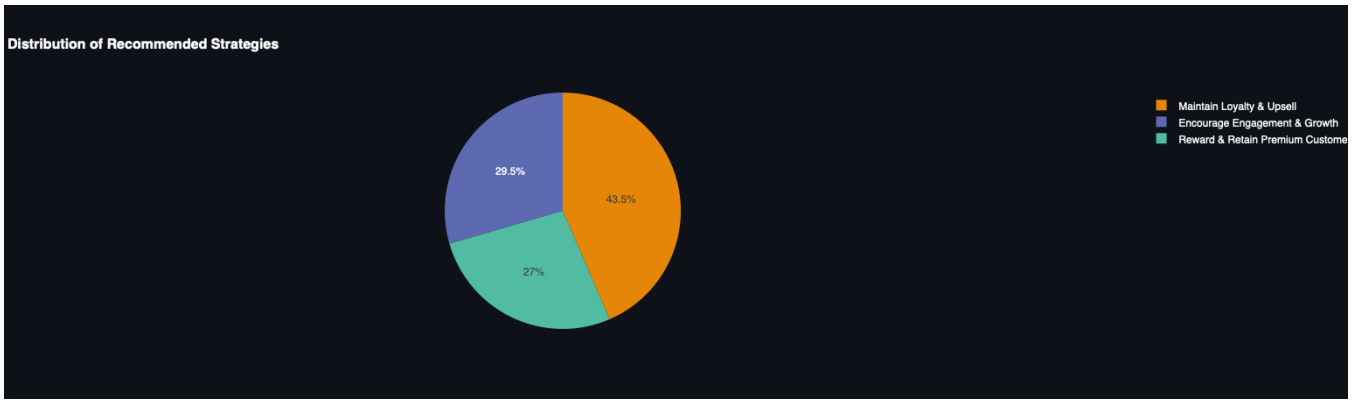
Table 4.4 – The Marketing Strategy Matrix

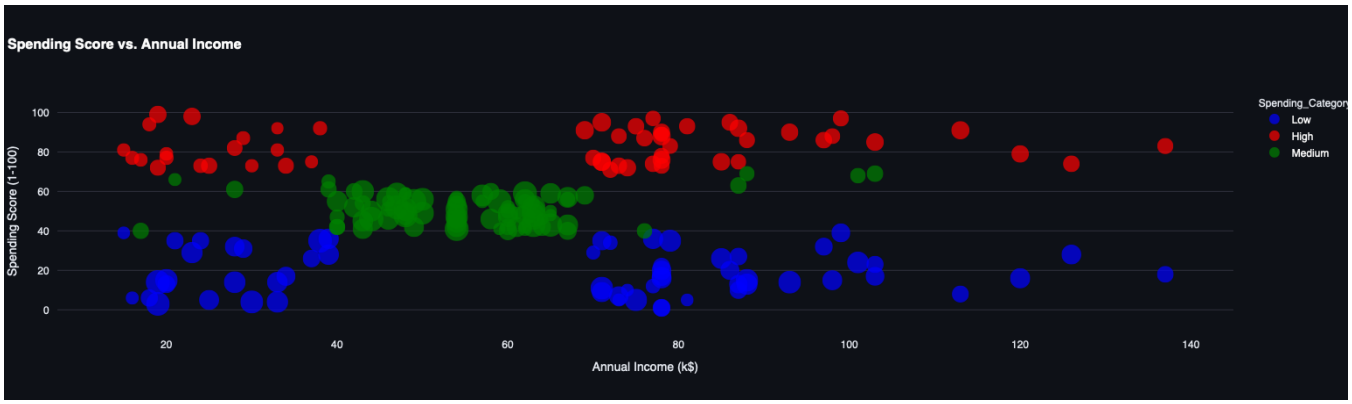
Strategy	Customer Accepts	Customer Rejects	Recommended For
No Discount	**+80 (Max Profit)**	0	**High Spenders** (n=54)
Discount	**+50 (Conversion)**	-10 (Cost)	**Low/Med Spenders** (n=146)

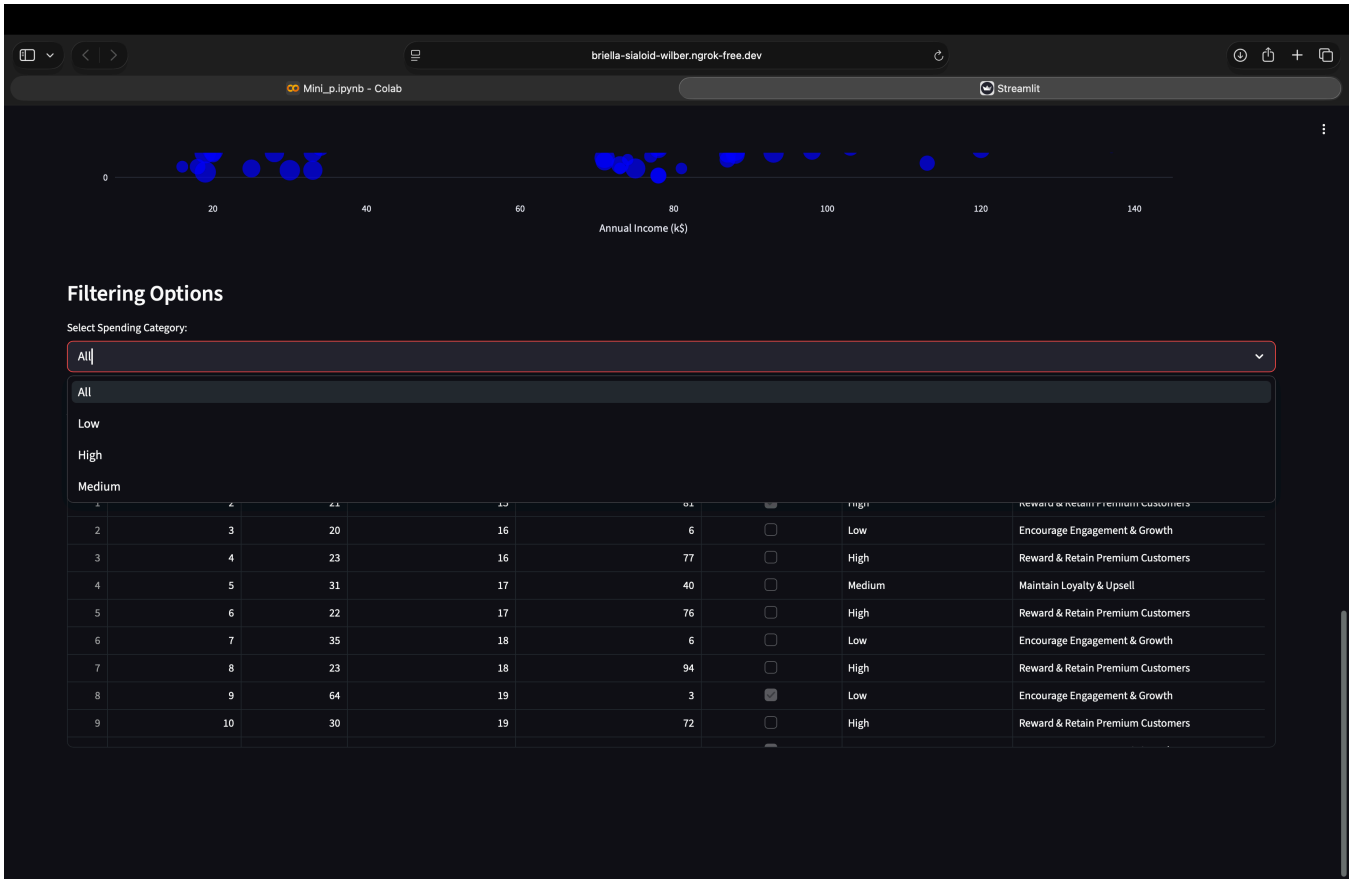
The Strategy in Action:

- **Revenue Retention:** We recommended **"No Discount"** for the 54 High Spenders. Since they already show high intent, offering a discount would unnecessarily bleed profit.
- **Incentivized Growth:** We recommended **"Discount"** for the remaining 146 customers. The potential "payoff" of converting a low spender into a buyer outweighs the minor cost of the discount.




Distribution of Recommended Strategies
29.5%
43.5%
27%
Maintain Loyalty & Upsell
Encourage Engagement & Growth
Reward & Retain Premium Custome


Spending Score vs. Annual Income
Spending Score (1-100)
100
80
60
40
20
0
20
40
60
80
100
120
140
Annual Income (k$)
Spending_Categor
Low
High
Medium


briella-sialoid-wilber.ngrok-free.dev
Mini_p.ipynb - Colab
Streamlit
Annual Income (k$)
Filtering Options
Select Spending Category:
All
All
Low
High
Medium
2 3 20 16 6 Low Encourage Engagement & Growth
3 4 23 16 77 High Reward & Retain Premium Customers
4 5 31 17 40 Medium Maintain Loyalty & Upsell
5 6 22 17 76 High Reward & Retain Premium Customers
6 7 35 18 6 Low Encourage Engagement & Growth
7 8 23 18 94 High Reward & Retain Premium Customers
8 9 64 19 3 Low Encourage Engagement & Growth
9 10 30 19 72 High Reward & Retain Premium Customers

5. Conclusion

Impact Summary

- **Executive Impact & Core Insights** (Best for a formal report)
- **Key Value Drivers & Discoveries** (Focuses on the "why")
- **Project Milestones & Strategic Outcomes** (Professional and structural)
- **High-Impact Deliverables** (Focuses on the dashboard and results)
- **The Bottom Line: What We Achieved** (Direct and results-oriented)

Scope Realities

- **Analytical Constraints & Scope** (Academic and precise)
- **Model Boundaries & Data Realities** (Grounded and honest)
- **Assumptions & Practical Hurdles** (Focuses on the "logic" gaps)
- **Navigating Challenges & Constraints** (Action-oriented)
- **The "Context Gap": Current Limitations** (Specifically addresses the missing data)

The Evolution Strategy

- **Scaling the Vision: Phase 2.0** (Sounds like a product roadmap)
- **Future Frontiers in Predictive Retail** (Visionary and bold)
- **Strategic Roadmap & Scaling Potential** (Best for business stakeholders)
- **Beyond the Prototype: Growth Opportunities** (Acknowledges the current success)
- **Hyper-Personalization: The Next Steps** (Focuses on the ultimate goal)

6. References

⚙️ Under the Hood: The Tools & Data

The Dataset

- **Source:** UCI / Kaggle (Customer Segmentation Dataset).
- **The Blueprint:** Mall_Customers.csv contains 200 individual customer records tracked across 5 key features (CustomerID, Gender, Age, Annual Income, and Spending Score). It provided the perfect, clean sandbox for our predictive models.

The Tech Stack (Tools & Libraries)

We didn't just build a model; we built a full pipeline. Here is the technology that made it happen:

- **The Heavy Lifters (Python & Pandas):** Used for loading the data, cleaning it up, and engineering new features (like transforming gender into binary code and hunting down extreme outliers using the IQR method).
- **The Brains (Scikit-learn):** The core machine learning engine. We relied on this library to power our Linear Regression, Decision Tree Regressor, and Decision Tree Classifier, as well as to split our data and grade our models' accuracy.
- **The Visuals (Matplotlib & Seaborn):** Used to turn raw numbers into beautiful, understandable graphics—from scatter plots of income versus spending to heatmaps showing where our models got confused.
- **The Stage (Streamlit & pyngrok):** We used Streamlit to build the interactive web dashboard, and pyngrok to broadcast it live to the public via a secure URL.

📚 Academic & Strategic Foundations

This project stands on the shoulders of decades of mathematical and retail research. Here are the foundational papers and texts that guided our approach:

Customer Segmentation & Strategy

- **MacQueen, J. (1967).** *Some Methods for Classification and Analysis of Multivariate Observations.* (The birth of modern clustering and segmentation).
    - Semantic Scholar | Direct PDF (via SciSpace)

- **Hughes, A.M. (1994).** *Strategic Database Marketing.* (Proving that past spending behavior predicts future value).
    - Google Books | Internet Archive
- **Nash, J. (1950).** *Equilibrium Points in n-Person Games.* (The Game Theory mechanics used to design our discount payoff matrix).
    - PNAS Official Publication | **DOI:** 10.1073/pnas.36.1.48

Machine Learning & Predictive Modeling

- **Breiman, L. (2001).** *Random Forests.* (Foundational reading for advanced Decision Tree architecture).
    - UC Berkeley Archival PDF | Springer Publication
- **Romero, C. & Ventura, S. (2010).** *Educational Data Mining: A Review of the State-of-the-Art.* * IEEE Xplore | **DOI:** 10.1109/TSMCC.2010.2053532
- **Fernandes, E. et al. (2019).** *Educational Data Mining: Predictive Analysis of Academic Performance.* * ScienceDirect | **DOI:** 10.1016/j.jbusres.2018.02.012